

一元积分学

1.华里士公式: $\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2}, \text{当} n \text{为偶数} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{2}{3} \cdot 1, \text{当} n \text{为奇数} \end{cases}$

2.定积分简化: $\begin{cases} \text{奇偶性简化:} \int_{-l}^l f(x) dx = \begin{cases} 0, \text{当} f(x) \text{为奇函数} \\ 2 \int_0^l f(x) dx, \text{当} f(x) \text{为偶函数} \end{cases} \\ \text{周期性简化:} \int_a^{a+T} f(x) dx = \int_0^T f(x) dx = \int_{\frac{T}{2}}^{\frac{T}{2}+T} f(x) dx \\ \text{几何意义简化:} \int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \pi a^2 \\ \text{正弦简化:} \int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx \end{cases}$

3.积分公式: $\begin{cases} \text{常用} x^k \text{类:} \int \frac{1}{x^2} dx = -\frac{1}{x} + C; \int \frac{1}{x} dx = \ln|x| + C; \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C; \\ \text{三角类:} \begin{cases} \int \frac{1}{\cos^2 x} dx = \int \sec^2 x dx = \tan x + C; \int \frac{1}{\sin^2 x} dx = \int \csc^2 x dx = -\cot x + C \\ \int \frac{1}{\cos x} dx = \int \sec x dx = \ln|\sec x + \tan x| + C; \int \csc x dx = \ln|\csc x - \cot x| + C \\ \int \sec x \tan x dx = \sec x + C; \int \csc x \cot x dx = -\csc x + C \\ \int \tan x dx = -\ln|\cos x| + C; \int \cot x dx = \ln|\sin x| + C \end{cases} \\ \text{反三角类} \begin{cases} \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C; \int \frac{dx}{1+x^2} = \arctan x + C \\ \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C; \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \end{cases} \\ \text{对数类:} \begin{cases} \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C; \int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C \\ \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C \end{cases} \end{cases}$

4.变限求导: $\begin{cases} \text{原始:} \left(\int_{\phi(x)}^{\phi(x)} f(t) dt \right)'_x = f[\phi(x)] \phi'(x) - f[\phi(x)] \bullet \phi'(x) \\ \text{化为纯} t \begin{cases} \int_0^x f(x-t) dt = \int_0^x f(u) du \\ \int_0^x (x-t) f(t) dt = x \int_0^x f(t) dt - \int_0^x t f(t) dt \\ \int_0^x t f(x-t) dt = x \int_0^x f(t) dt - \int_0^x t f(t) dt \\ \int_0^1 f(xt) dt = \frac{\int_0^x f(u) du}{x} (x \neq 0) \end{cases} \end{cases}$

5.积分嵌套: $\begin{cases} \text{一元:} f(x) = g(x) + \int_a^b f(x) dx, \begin{cases} \text{令} A = \int_a^b f(x) dx, \text{进而有} \\ A = \int_a^b (g(x) + A) dx \end{cases} \\ \text{二元:} f(x, y) = g(x, y) + \iint_D f(x, y) d\sigma, \begin{cases} \text{令} A = \iint_D f(x, y) d\sigma, \text{进而有} \\ A = \iint_D (g(x, y) + A) d\sigma \end{cases} \end{cases}$

6.区间再现: $\begin{cases} \text{基本理论:} \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \\ \text{示例:} \begin{cases} \int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx \\ \int_{\frac{\pi}{2}}^{\frac{\pi}{2}+e^x} \frac{\sin^4 x}{1+e^x} dx = \frac{3\pi}{16} \end{cases} \end{cases}$

7.分部积分: $\begin{cases} \text{公式:} \int u dv = uv - \int v du \\ \text{几种情况} \begin{cases} \text{①对变限积分函数再积分, 用分部积分} \\ \text{②对形如} x f' \text{的函数积分, 用分部积分} \\ \text{③对含对数或反三角的函数积分, 用分部积分} \quad \begin{cases} \text{例:} \int x \ln x dx \\ \text{例:} \int \arctan x dx \end{cases} \\ \text{④对形如} x e^x, x \sin x, x \cos x \text{的函数积分, 用分部积分} \end{cases} \end{cases}$

8.积分中值: $\begin{cases} \text{①函数平均值:} \bar{f} = \frac{\int_a^b f(x) dx}{b-a} \\ \text{②积分中值定理:} \int_a^b f(x) dx = f(\xi)(b-a) (\text{要求:} f \text{连续}) \\ \text{③推广的积分中值:} \int_a^b f(x) g(x) dx = f(\xi) \int_a^b g(x) dx (\text{要求:} f, g \text{连续, 且} g \text{不变号}) \\ \text{示例:} \lim_{n \rightarrow \infty} \int_0^1 x^n f(x) dx = 0 (\text{其中} f(x) \text{连续}) \end{cases}$

9.积分换元法: $\begin{cases} \text{三角代换:} \begin{array}{|c|c|c|c|} \hline \text{标志} & \text{代换} & \sin t & \cos t \\ \hline \sqrt{a^2 - x^2} & x = a \sin t & \frac{x}{a} & \frac{\sqrt{a^2 - x^2}}{a} \\ \hline \sqrt{a^2 + x^2} & x = a \tan t & \frac{x}{\sqrt{a^2 + x^2}} & \frac{a}{\sqrt{a^2 + x^2}} \\ \hline \sqrt{x^2 - a^2} & x = a \sec t & \frac{\sqrt{x^2 - a^2}}{x} & \frac{a}{x} \\ \hline \end{array} \\ \text{其他代换} \begin{cases} \text{①根式代换:根式中无平方, 用根式整体代换} \\ \text{示例:} \int \frac{dx}{\sqrt{1+e^x}} = \int \frac{2dt}{t^2-1} = 2 \ln(\sqrt{1+e^x}-1) - x + c \\ \text{②倒代换:分母分子阶次差2阶及以上, 用倒代换} \\ \text{示例:} \int \frac{dx}{x(1+x^2)} = \int \frac{-tdt}{t^2+1} = -\frac{1}{2} \ln(1+\frac{1}{x^2}) + c \end{cases} \end{cases}$

参比积分: $\begin{cases} \int_0^1 \frac{1}{x^p} dx \begin{cases} 0 < p < 1 \text{时, 收敛,} \\ p \geq 1 \text{时, 发散,} \end{cases} \\ \int_1^{+\infty} \frac{1}{x^p} dx \begin{cases} p > 1 \text{时, 收敛,} \\ p \leq 1 \text{时, 发散,} \end{cases} \end{cases}$

10.反常积分: $\begin{cases} \text{审敛法:} \begin{cases} \text{被积函数无穷小时, } \lim_{x \rightarrow +\infty} \frac{f(x)}{\frac{1}{x^p}} = l: \begin{cases} \text{若} 0 \leq l < +\infty, p > 1, \text{则} \int_a^{+\infty} f(x) dx \text{收敛} \\ \text{若} 0 < l \leq +\infty, p \leq 1, \text{则} \int_a^{+\infty} f(x) dx \text{发散} \end{cases} \\ \text{被积函数无穷大时, } \lim_{x \rightarrow a^+} \frac{f(x)}{\frac{1}{(x-a)^p}} = l: \begin{cases} \text{若} 0 \leq l < +\infty, p < 1, \text{则} \int_a^b f(x) dx \text{收敛} \\ \text{若} 0 < l \leq +\infty, p \geq 1, \text{则} \int_a^b f(x) dx \text{发散} \end{cases} \end{cases} \end{cases}$

$\int_2^{+\infty} \frac{1}{x^m \ln^n x} dx$ 的敛散性($m, n > 0$) $\begin{cases} m > 1, \text{必收敛} \\ m = 1 \begin{cases} n > 1, \text{收敛} \\ n \leq 1, \text{发散} \end{cases} \\ m < 1, \text{必发散} \end{cases}$

1.基本有理函数的原函数: $\begin{cases} \int \frac{A}{x-a} dx = A \ln|x-a| + C \\ \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C; \int \frac{dx}{1+x^2} = \arctan x + C \\ \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C; \int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C \end{cases}$

11.有理函数: $\begin{cases} \text{假分式: 通过观察法或长除法分解为函数+真分式} \\ \text{2.有理函数的拆分} \begin{cases} \text{按照分母中的因子进行分解} \\ \text{真分式的分解} \begin{cases} \text{求系数的方法:} \begin{cases} \text{待定系数法} \\ \text{留数法(或暇点法)} \end{cases} \end{cases} \end{cases} \\ \text{3.分解示例} \begin{cases} \frac{1}{x(x^2-1)(x+1)} = \frac{1}{x(x-1)(x+1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} + \frac{D}{(x+1)^2} \\ \text{其中} A = -1, B = \frac{1}{4}, C = \frac{3}{4}, D = \frac{1}{2} \end{cases} \end{cases}$

12.定积分几何应用: $\begin{cases} \text{面积微元:} \begin{cases} \text{直角坐标:} dA = y dx, \text{或} dA = x dy \\ \text{极坐标:} dA = \frac{1}{2} r^2 d\theta \end{cases} \\ \text{体积微元:} dV = \pi r^2 \cdot dx (\text{圆盘}), dV = 2\pi r \cdot y \cdot dx (\text{柱壳}) \\ \text{弧微元:} \begin{cases} ds = \sqrt{(x')^2 + (y')^2} dt (\text{参数方程}) \\ ds = \sqrt{1+y'^2} dx (\text{直角坐标}) \\ ds = \sqrt{r^2 + r'^2} d\theta (\text{极坐标}) \end{cases} \\ \text{表面积微元:} dS = 2\pi r \cdot ds (\text{数三不要求}) \end{cases}$

13.形心坐标: $\begin{cases} \bar{x} = \frac{\iint_D x dx dy}{\iint_D dx dy} (\text{平面}); \bar{x} = \frac{\iiint_D x dS}{\iint_D dS} (\text{曲面, 仅数一}); \bar{x} = \frac{\iiint_D x dV}{\iiint_D dV} (\text{立体, 仅数一}) \\ \text{已知形心, 可简化积分, 如:} \iiint_D x dx dy = \bar{x} \cdot \iint_D dx dy (\text{其他同理}) \end{cases}$

14.两类三角积分: $\begin{cases} \text{降幂:} \int \sin^4 x dx = \frac{1}{8} \int (3 - 4 \cos 2x + \cos 4x) dx \\ \text{凑微分:} \int \sin^2 x dx = -\int (1 - \cos^2 x)^2 d \cos x \end{cases}$

15.可积性: $\begin{cases} \text{必要条件: 若} \int_a^b f(x) dx \text{必定存在, } f(x) \text{在} [a, b] \text{上有界} \\ \text{充分条件: 若} f(x) \text{在} [a, b] \text{上} \begin{cases} \text{连续} \\ \text{有界, 且只有有限个间断点} \\ \text{只有有限个第一类间断点} \\ \text{单调有界} \end{cases}, \text{则} \int_a^b f(x) dx \text{必定存在} \end{cases}$

16.原函数存在定理: $\begin{cases} 1. \text{连续函数一定存在原函数.} \\ 2. \text{存在第一类间断点的函数一定不存在原函数.} \\ \text{但在不含第一类间断点的子区间可能存在原函数 (子区间连续时, 必存在原函数).} \\ 3. \text{存在第二类间断点的函数可能存在与原函数.} \end{cases}$

17.定积分物理应用(数三不考) $\begin{cases} \text{细棒的质心:} \bar{x} = \frac{\int_a^b x dm}{\int_a^b dm} = \frac{\int_a^b x \rho(x) dx}{\int_a^b \rho(x) dx} \\ \text{变力做功(沿直线):} dW = F \cdot dx, W = \int_a^b F \cdot dx \\ \text{抽水做功:} dW = dm \cdot g \cdot h(x) = \rho g h dV = \rho g h(x) A(x) dx, W = \int_a^b \rho g h(x) A(x) dx \\ \text{液体侧压力:} dF = PdS = \rho g h dS \\ \text{万有引力:} dF = \frac{GMdm}{r^2} (dF_x = \frac{GMdm}{r^2} \cos \theta, dF_y = \frac{GMdm}{r^2} \sin \theta) \end{cases}$